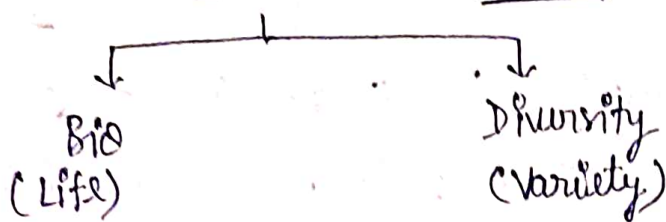


# "BIO-DIVERSITY"



"The term biodiversity refers to the variety of living species on Earth, including plants, animals, bacteria and fungi."

Flora → Plants, Herbs, trees

Fauna → Animals, bacteria, micro-organisms

} Combination of this known as "Bio-diversity".

## Types of Bio-diversity

### Genetic diversity

"is the variation in genes and genotypes within a species."

E.g. Every human look different from each other

### Species

"is the variety of species within a habitat or a region."

E.g.

Fish

Delphin Sarc etc.

### Ecosystem

"refers to the variations in the plant and animal species living together and connected by food chains and food web."

E.g. Desert  
Pond  
Forest Rain

## Importance of Bio-diversity

- (01) Integral part of cultural identity
- (02) Used for medicinal purpose
- (03) Contributes to climate stability
- (04) Provide more food resources
- (05) Relation b/w Bio-diversity & industry
- (06) Economic Benefits of Bio-div.
- (07) Helps in Pollution Management
- (08) Forming a Healthy Ecosystem

## ⇒ In-situ & Ex-situ Conservation of Biodiversity

### In-situ (onsite) conservation

① species are conserved in their natural habitat. (safe from man).

② Applicable for large populations

③ Require large area

④ Less expensive

⑤ provide greater mobility to the species because of large area.

Ex: National Park, Biosphere reserves, sanctuaries

### Ex-situ (offsite) conservation

species are conserved in outside the natural habitat.

(Conservation outside from natural habitat.)

Applicable for small population

Require less area

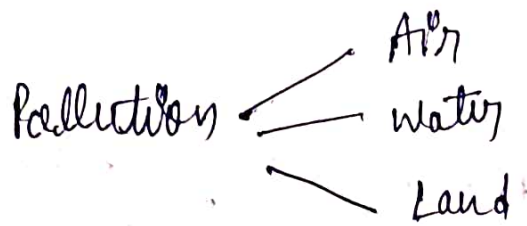
More expensive

provide less mobility to the organisms because of small area.

Ex: Aquarium, zoological park, Botanical garden

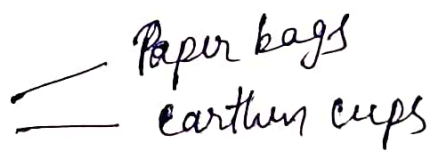
# CO<sub>2</sub> Pollution

is: Introduction of harmful materials into the environment.  
These harmful materials are called pollutants.



⇒ Role of individual in prevention of pollution

- (i) Develop respect of nature
- (ii) Reduce the use of wood & paper product
- (iii) Try to recycle paper product
- (iv) Use public transport and for a short distance use bicycle or go on foot.
- (v) Practice organic farming by using biofertilizer & biopesticides instead of inorganic chemicals
- (vi) Always use ecofriendly product
- (vii) Cut down the use of CFCs.
- (viii) Don't put harmful chemical into drain or H<sub>2</sub>O bodies.
- (ix) Protect trees from cutting
- (x) Use rechargeable batteries to reduce metal pollution
- (xi) Plant more trees in your area
- (xii) Don't throw wrapper on road bring it back home with you.





## ⇒ Human population

A population is defined as a group of individuals of the same species living and interbreeding within a given area.  
↓  
Growth of family

## Population density

- (i) The no. of individuals living within that specific location determines the population density, or the number of individual divided by the size of the area.
- (ii) Population density can be used to describe the location, growth and migration of many organisms.